

MOHANRAJ VELMANI

AI ENGINEER | MACHINE LEARNING ENGINEER | DJANGO & MLOPS DEVELOPER

CONTACT

- 9360239398
- mohanrajvmvm221023@gmail.com
- [linkedin](#)
- [GitHub](#)
- [Portfolio](#)

PROFILE SUMMARY

AI Engineer with strong expertise in Machine Learning, Deep Learning, NLP, and MLOps, backed by hands-on experience in Django REST APIs, PyTorch, Scikit-learn, and production deployment using Docker and cloud platforms. Proven ability to design end-to-end AI systems—from data engineering and feature extraction to model training, evaluation, deployment, and monitoring. Experienced in building real-world AI products such as churn prediction systems, image classification pipelines, meeting transcription & diarization tools, and time-series forecasting solutions.

EDUCATION

AVS COLLEGE OF ARTS AND SCIENCE

2020 - 2023

- Computer Science

SKILLS

Artificial Intelligence & Machine Learning

- Supervised & Unsupervised Learning
- Regression & Classification Models
- Clustering (KMeans, DBSCAN)
- Dimensionality Reduction (PCA, t-SNE)
- Feature Engineering & Feature Selection
- Model Evaluation & Hyperparameter Tuning

Deep Learning

- Neural Networks & CNNs
- Image Classification (MNIST, CIFAR-10)
- Transfer Learning & ResNet Architectures
- PyTorch (Model building, training loops, optimizers)

Natural Language Processing

- Speech-to-Text (Whisper - local models)
- Audio Transcription & Sentiment Analysis
- Text Classification & Embeddings
- Hugging Face Models

Programming & Frameworks

- Python (Advanced)
- Django & Django REST Framework
- FastAPI
- Pydantic

MLOps & Deployment

- Docker & Containerization
- Model Packaging & Artifacts
- REST API-based ML Serving
- Render
- Experiment Tracking (Weights & Biases)
- CI/CD Basics

Data & Tools

- NumPy, Pandas
- Scikit-learn
- XGBoost
- SQL & SQLite

PROFESSIONAL EXPERIENCE

Zoho Creator Developer - IT Executive OCT 2024 - PRESENT

- Developed and maintained low-code business applications for enterprise workflows.
- Migrated Zoho Creator applications to custom Django + React.js architectures.
- Designed backend logic, APIs, and database models.
- Collaborated with stakeholders to optimize performance and scalability.
- Gained strong understanding of business automation and production-grade systems.

KEY AI ML PROJECTS

Customer Churn Prediction System

- Developed a full-stack ML system using Django + Scikit-learn.
- Feature engineering on customer behavioral data.
- Implemented REST API for real-time predictions.
- Dockerized and deployed to Render.
- Production-ready structure with model versioning.

Image Classification with PyTorch

- Implemented CNNs from scratch for MNIST & CIFAR-10.
- Explained pixel-level learning and backpropagation.
- Implemented ResNet & transfer learning.
- Integrated Weights & Biases for experiment tracking.

AI Meeting Assistant

- Built an AI-powered meeting assistant using Streamlit.
- Implemented speech-to-text using Whisper (local models).
- Added speaker diarization and sentiment analysis.
- Processed real-world audio with noise and overlapping speakers.

ACHIEVEMENTS & STRENGTHS

- Strong problem-solving and system design mindset
- Excellent understanding of how models learn internally
- Ability to convert business problems into AI solutions
- Production-focused AI engineering approach
- Fast learner with strong self-driven research capability